

Module 3:

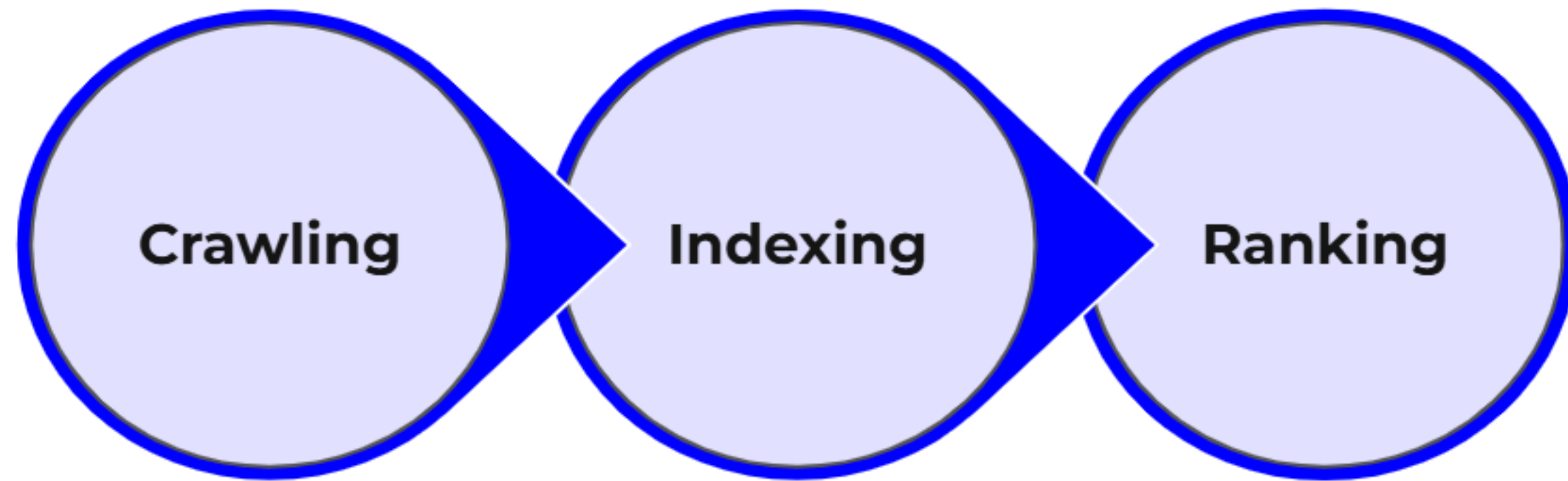
Search Engines



Topic 1: How Do Search Engines Work?



How do Search Engines Work?



How do Search Engines Work?

Crawling is the process through which the search engine visits and discovers the contents of the web pages by using automated bots called web crawlers (spiders).

Indexing is the process of analyzing the data collected in the crawling process and storing it in databases to be eligible to show up in search results.

Ranking is the order in which the indexed results appear in the results pages according to some ranking algorithms.

How do Search Engines Work?

robots.txt (robots exclusion protocol) is a set of instructions for web crawlers that informs them how to crawl pages on a website.

Directive/	Usage
User-agent:	Specifies the web crawler
Allow:	Specifies a directory or page that you want the user agent to crawl
Disallow:	Specifies a directory or page that you don't want the user agent to crawl
Sitemap:	Specifies the location of the sitemap file

Notes:

- **#** means the beginning of a comment.
- ***** is a wildcard used to represent anything. **For example:**

User-agent: *
Disallow:



Allow all crawlers complete access

robots.txt and sitemap.xml

sitemap.xml is an XML file that lists the URLs for a site, and it allows the site to be crawled more efficiently.

sitemap.xml => URL inclusion protocol

robots.txt => URL exclusion protocol

<https://www.google.com/sitemap.xml>

<https://www.nytimes.com/robots.txt>